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APEX RETAIL GROUP | ANALYTICS TRAINING PROGRAM
DOCUMENT: CLASS NOTES (SESSION 1)
TOPIC: THE FIRST TICKET — UNDERSTANDING BUSINESS QUESTIONS & COMPUTATIONAL LOGIC
LEVEL: BEGINNER (DAY 1)

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1. BUSINESS SCENARIO: YOUR FIRST DAY AT APEX RETAIL GROUP

Welcome to Apex Retail Group. You are joining the business as a Junior Data Analyst. Your primary role is not to write code for the sake of coding; it is to help business leaders answer critical questions using evidence.

On your very first morning, Sarah Jenkins, the HR Operations Manager, stops by your desk with an urgent request:

> "We are setting up a specialized Analytics & Insights team under Apex Retail. I have a preliminary list of five newly hired analysts and their proposed base salaries. Before I submit this compensation package to the VP of HR, I need a quick baseline review:
> **1. What is the average base salary for this new team?**
> **2. How does each individual salary compare to our corporate standard baseline salary band of $75,000?**
> Can you run these numbers quickly and give me a brief summary?"

As a Junior Data Analyst, your first instinct should NEVER be to randomly press buttons in software or write unstructured code. Your job is to process the request through a structured analytical workflow:

Business Request ⟶ Analytical Questions ⟶ Data Inspection ⟶ Tool Selection ⟶ Analytical Calculation ⟶ Business Recommendation

2. TRANSLATING BUSINESS REQUESTS INTO ANALYTICAL QUESTIONS

Stakeholders speak in business language ("quick review", "how does it look?", "are we spending too much?"). Analysts translate business language into precise mathematical and analytical questions:

Stakeholder Business Request	Translated Analytical Question
"What is the average base salary?"	What is the arithmetic mean of the five salary values in the dataset?
"How does each salary compare to $75,000?"	For each individual salary, what is the absolute dollar difference: (Salary - $75,000)?

3. COMPUTATIONAL THINKING: HOW COMPUTERS STORE AND PROCESS DATA

To answer Sarah's question, we need a digital environment to store numbers and perform operations. We use Python inside a Jupyter Notebook environment.

A. Storing Single Data Points: Variables and Data Types

In analysis, data points represent real-world entities. To track a single analyst's salary, we assign a value to a named container called a VARIABLE.

Examples of fundamental data types in Python:

- **Integer (int):** Whole numbers.
  Example: `employee_id = 101`
- **Float (float):** Numbers with decimal places (used for precision currency/metrics).
  Example: `salary = 78500.00`
- **String (str):** Text characters wrapped in quotes.
  Example: `employee_name = "Alex Chen"`
- **Boolean (bool):** Truth values (True or False).
  Example: `is_above_baseline = True`

B. Storing Collections of Data: Lists

Sarah provided five salaries: `$72,000, $81,000, $75,000, $69,000,` and `$88,000.` Creating five separate variable names (salary1, salary2, etc.) is inefficient and breaks down when managing thousands of rows.

Instead, we group related data into a sequence called a LIST. In Python, lists are created using square brackets `` `[]` `` with comma-separated elements:

```
salaries = [72000.0, 81000.0, 75000.0, 69000.0, 88000.0]
```

4. EXECUTING THE ANALYSIS IN PYTHON

Below is the clean, annotated Python code required to perform Sarah's salary review:

```python
# Step 1: Define the baseline standard salary band
baseline_salary = 75000.0

# Step 2: Store the proposed team salaries in a list
team_salaries = [72000.0, 81000.0, 75000.0, 69000.0, 88000.0]

# Step 3: Calculate total payroll and total number of team members
total_payroll = sum(team_salaries)
team_count = len(team_salaries)

# Step 4: Calculate arithmetic average (mean)
average_salary = total_payroll / team_count

# Step 5: Evaluate salary gap against baseline
# Positive gap = Above baseline; Negative gap = Below baseline
salary_gaps = [
    team_salaries[0] - baseline_salary,
```

```
    team_salaries[1] - baseline_salary,
    team_salaries[2] - baseline_salary,
    team_salaries[3] - baseline_salary,
    team_salaries[4] - baseline_salary
]

# Step 6: Output results
print("--- HR SALARY REVIEW RESULTS ---")
print("Total Payroll Expense: $", total_payroll)
print("Team Member Count:     ", team_count)
print("Average Base Salary:   $", average_salary)
print("Individual Gaps:       ", salary_gaps)
```

INTERPRETING NUMBERS & FORMULATING BUSINESS RECOMMENDATIONS

A data analyst NEVER hands raw output or code directly to a manager. You must translate raw mathematical outputs into business insights and recommendations.

A. Observation (The Raw Data):

- Total Team Payroll: $385,000.00 across 5 hires.
- Average Salary: $77,000.00.
- Individual Variances from $75,000 baseline:
    - Hire 1: -$3,000.00 (Below baseline)
    - Hire 2: +$6,000.00 (Above baseline)
    - Hire 3: $0.00 (On baseline)
    - Hire 4: -$6,000.00 (Below baseline)
    - Hire 5: +$13,000.00 (Significantly above baseline)

B. Interpretation & Business Meaning:

- The team's average salary ($77,000) exceeds the corporate baseline benchmark ($75,000) by $2,000 per employee (+2.67%).
- The overall budget overrun is driven primarily by Hire 5 ($88,000), who sits $13,000 above the standard band.
- Two entry-level hires (Hire 1 at $72k, Hire 4 at $69k) offset some of this premium, keeping the total department payroll budget within a reasonable range.

C. Professional Recommendation to Sarah Jenkins (HR Operations Manager):

"The proposed total compensation for the 5-member Analytics team is $385,000, yielding an average salary of $77,000—slightly above our standard $75,000 benchmark. This +2.67% variance is heavily driven by a single senior hire at $88,000. Recommendation: If the $88,000 offer reflects specialized senior skills, approve the package as the lower entry rates ($69k–$72k) keep total departmental variance to a manageable $10,000 over baseline."

CORE MINDSET HABITS REINFORCED TODAY

1. Never stop at a number: A calculated mean of $77,000 is meaningless until it is compared against the $75,000 corporate benchmark.
2. Structure before syntax: Frame the question ("What is the variance?") before writing the Python code (`sum() / len()`).